

Design the interface circuit in Figure P3-67 so that $R_{IN} = 100\ \Omega$ and the current delivered to the $50\text{-}\Omega$ load is 50 mA . (Hint: Use an L-pad.)

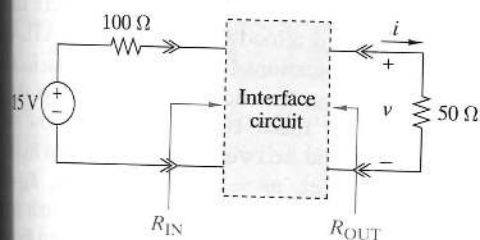


FIGURE P3-67

Design the interface circuit in Figure P3-67 so that $R_{IN} = 50\ \Omega$ and the voltage delivered to the $50\text{-}\Omega$ load is 3 V . (Hint: Use an L-pad.)

The circuit in Figure P3-69 has a source resistance of $75\ \Omega$ and a load resistance of $600\ \Omega$. Design the interface circuit so that the input resistance is $R_{IN} = 75\ \Omega \pm 10\%$ and the output resistance is $R_{OUT} = 600\ \Omega \pm 10\%$.

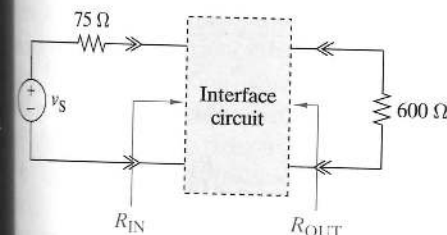


FIGURE P3-69

It is claimed that both interface circuits in Figure P3-70 will deliver $v = 4\text{ V}$ to the $75\text{-}\Omega$ load. Verify this claim. Which interface circuit consumes the least power? Which interface circuit has the output resistance that best matches the $75\text{-}\Omega$ load?

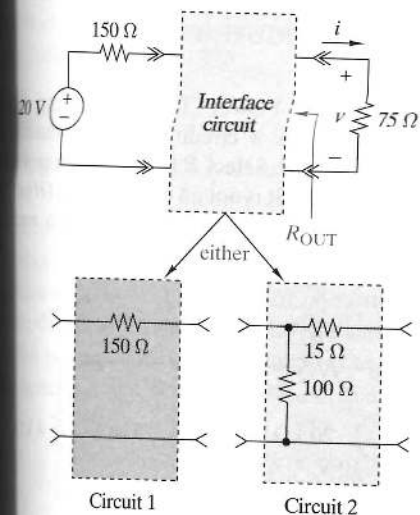
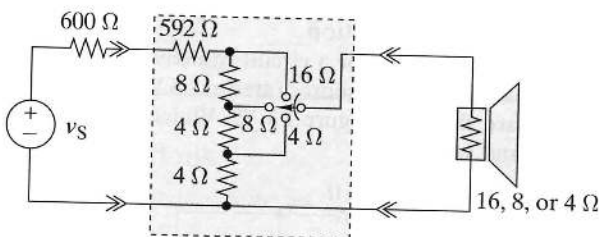


FIGURE P3-70

INTEGRATING PROBLEMS

3-71 Audio Speaker Resistance-Matching Network

A company is producing an interface network that they claim will result in R_{IN} of $600\ \Omega \pm 2\%$ and R_{OUT} of $16, 8, \text{ or } 4\ \Omega \pm 2\%$ —depending on whether the connected speakers are $16, 8, \text{ or } 4\ \Omega$ —selectable via a built-in switch. Prove or disprove their claim.

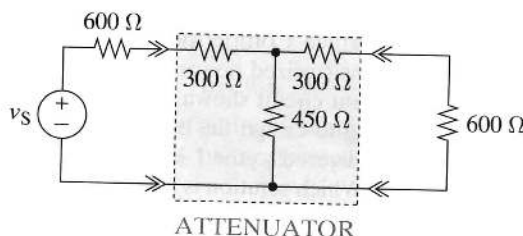


Audio matching network

FIGURE P3-71

3-72 Attenuation Analysis

In Figure P3-72 a two-port attenuator connects a $600\text{-}\Omega$ source to a $600\text{-}\Omega$ load. Find the power delivered to the load in terms of v_S . Remove the attenuator and find the power delivered to the load when the source is directly connected to the load. By what fraction does the attenuator reduce the power delivered to the $600\text{-}\Omega$ load? Express the fraction in dB.



ATTENUATOR

FIGURE P3-72

3-73 Interface Circuit Design

Using no more than three $50\text{-}\Omega$ resistors, design the interface circuit in Figure P3-73 so that $v \leq 4\text{ V}$ and $i \leq 100\text{ mA}$ regardless of the value of R_L .

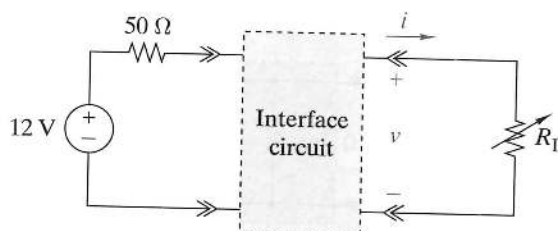


FIGURE P3-73

3-74 **Battery Design**

A satellite requires a battery with an open-circuit voltage $v_{OC} = 36\text{ V}$ and a Thévenin resistance $R_T \leq 10\ \Omega$. The battery is to be constructed using series and parallel combinations of one of two types of cells. The first type has $v_{OC} = 9\text{ V}$, $R_T = 4\ \Omega$, and a weight of 40 grams. The second type has $v_{OC} = 4\text{ V}$, $R_T = 0.5\ \Omega$, and a weight of 15 grams. Design a minimum-weight battery that meets the open-circuit voltage and Thévenin resistance requirements.

3-75 **Design Evaluation**

A requirement exists for a circuit to deliver 0 to 5 V to a $100\text{-}\Omega$ load from a 20-V source rated at 2.5 W. Two proposed circuits are shown in Figure P3-75. Which one would you choose and why?

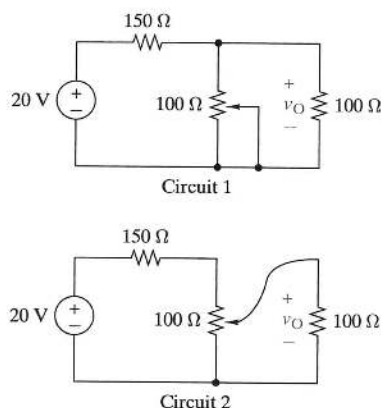


FIGURE P3-75

3-76 **Design Interface Competition**

The output of a transistorized power supply is modeled by the Norton equivalent circuit shown in Figure P3-76. Two teams are competing to design the interface circuit so that $25\text{ mW} \pm 10\%$ is delivered to the $1\text{-k}\Omega$ load resistor. Their designs are shown. Which solution is better, considering the use of standard values, number of parts, and power required by the source? Would your choice be different if the power had to be within $\pm 5\%$?

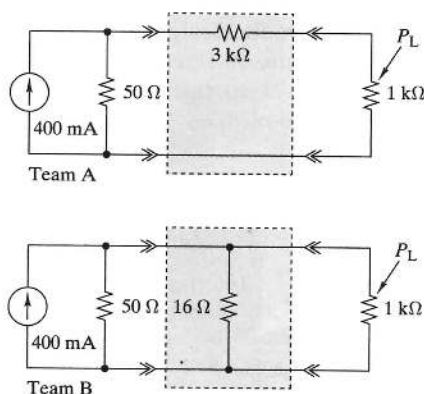


FIGURE P3-76

3-77 **Analysis of Competing Interfaces Using MATLAB**

Figure P3-77 displays two generalized interface circuits. In both circuits, resistors R_1 and R_2 connect the Thévenin equivalent circuit to a load resistor. Using MATLAB, develop symbolic expressions for the load current i_L and the input resistance, R_{IN} , for each circuit in terms of the circuit parameters. Using these two expressions, use the MATLAB command `solve` to solve for R_1 and R_2 in terms of i_L and R_{IN} . Let $v_T = 15\text{ V}$, $R_T = 100\ \Omega$, $i_L = 50\text{ mA}$, and $R_{IN} = 100\ \Omega$. Can you use these interface circuits to find suitable values for R_1 and R_2 that meet these specifications? Compare your interface circuit solution to the solution to Problem 3-67.

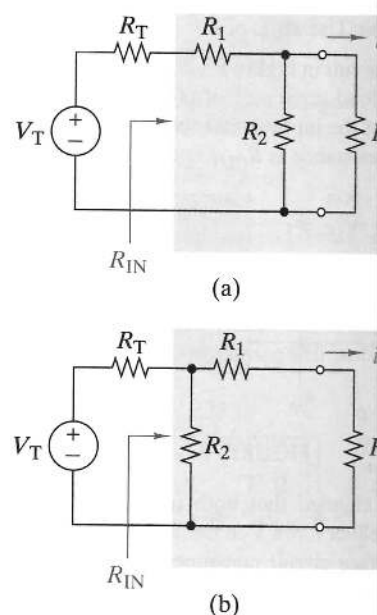


FIGURE P3-77

3-78 **Maximum Power Transfer Using MATLAB**

Figure P3-78 shows a circuit with two sources, a load resistor R , and a resistor R . Select R for maximum power transfer to the load. The result is not an obvious one. (Use MATLAB or OrCAD using the `PARAMETER` function to vary R from $0.1\ \Omega$ to $100\text{ M}\Omega$.)

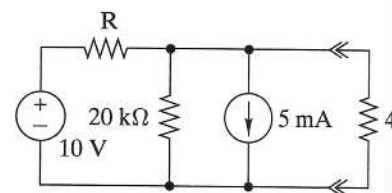


FIGURE P3-78

output voltage will be half of the Thévenin voltage, or $v_s/8$, because the load resistance matches the Thévenin resistance.

SUMMARY

- Node-voltage analysis involves identifying a reference node and the node to datum voltages at the remaining $N-1$ nodes. The KCL connection constraints at the $N-1$ nonreference nodes combined with the element constraints written in terms of the node voltages produce $N-1$ linear equations in the unknown node voltages.
- Mesh-current analysis involves identifying mesh currents that circulate around the perimeter of each mesh in a planar circuit. The KVL connection constraints around $E-N+1$ meshes combined with the element constraints written in terms of the mesh currents produce $E-N+1$ linear equations in the unknown mesh currents.
- Node and mesh analysis can be modified to handle both types of independent sources using a combination of three methods: (1) source transformations, (2) selecting circuit variables so independent sources specify the values of some of the unknowns, and (3) using supernodes or supermeshes.
- A circuit is linear if it contains only linear elements and independent sources. For single-input linear circuits, the proportionality property states that any output is proportional to the input. For multiple-input linear circuits, the superposition principle states that any output can be found by summing the output produced when each input acts alone.
- A Thévenin equivalent circuit consists of a voltage source in series with a resistance. A Norton equivalent circuit consists of a current source in parallel with a resistance. The Thévenin and Norton equivalent circuits are related by a source transformation.
- The parameters of the Thévenin and Norton equivalent circuits can be determined using any two of the following: (1) the open-circuit voltage at the interface, (2) the short-circuit current at the interface, and (3) the equivalent resistance of the source circuit with all sources turned off.
- The parameters of the Thévenin and Norton equivalent circuits can also be determined using circuit reduction methods or by directly solving for the source i - v relationship using node-voltage or mesh-current analysis.
- For a fixed source and an adjustable load, the maximum interface signal levels are $v_{\text{MAX}} = v_{\text{OC}} (R_L = \infty)$, $i_{\text{MAX}} = i_{\text{SC}} (R_L = 0)$, and $p_{\text{MAX}} = v_{\text{OC}} i_{\text{SC}} / 4 (R_L = R_T)$. When $R_L = R_T$, the source and load are said to be matched.
- Interface signal transfer conditions are specified in terms of the voltage, current, or power delivered to the load. The design constraints depend on the signal conditions specified and the circuit parameters that are adjustable. Some design requirements may require a two-port interface circuit. An interface design problem may have one, many, or no solutions.

PROBLEMS

OBJECTIVE 3-1 GENERAL CIRCUIT ANALYSIS (SECT. 3-1 TO 3-2)

Given a circuit consisting of linear resistors and independent sources:

- (Formulation) Write node-voltage or mesh-current equations for the circuit.
 - (Solution) Solve the equations from part (a) for selected signal variables or input-output relationships using classical or software computational techniques.
- Node-voltage method: See Examples 3-1, 3-2, 3-3, 3-4, 3-5, 3-6 and Exercises 3-2, 3-3, 3-4, 3-5, 3-6, 3-7, 3-8, 3-9, 3-10
 Mesh-current method: See Examples 3-7, 3-8, 3-9 and Exercises 3-11, 3-12, 3-13, 3-14, 3-15, 3-16

3-1 (a) Formulate node-voltage equations for the circuit in Figure P3-1.

- Solve these equations for v_A and v_B .
- Use these results to find v_x and i_x .

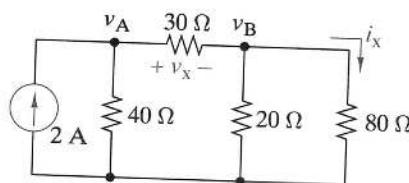


FIGURE P3-1

- 3-2 (a) Formulate node-voltage equations for the circuit in Figure P3-2.
 (b) Solve these equations for v_A and v_B .
 (c) Use these results to find v_x and i_x .

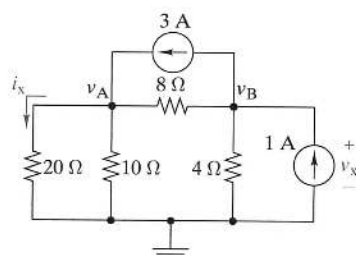


FIGURE P3-2

- 3-3 (a) Formulate node-voltage equations for the circuit in Figure P3-3.

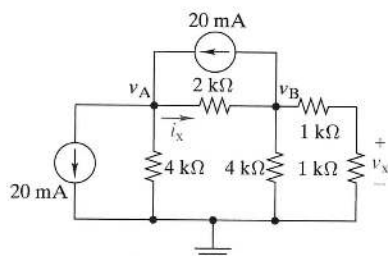


FIGURE P3-3

- (b) Solve these equations for v_A and v_B .
(c) Use these results to find v_x and i_x .

- 3-4 (a) Formulate node-voltage equations for the circuit in Figure P3-4.

- (b) Solve these equations for v_A and v_C .
(c) Use these results to find v_x and i_x .

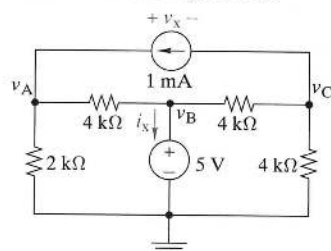


FIGURE P3-4

- 3-5 (a) Choose a ground wisely and formulate node-voltage equations for the circuit in Figure P3-5.

- (b) Solve for v_x and i_x when $R_1 = R_2 = R_3 = R_4 = 1 \text{ k}\Omega$, $v_S = 20 \text{ V}$, and $i_S = 1 \text{ mA}$.

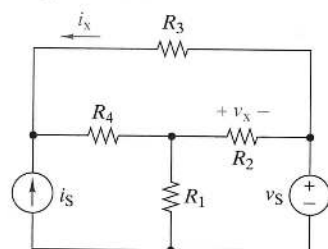


FIGURE P3-5

- 3-6 (a) Formulate node-voltage equations for the circuit in Figure P3-6.

- (b) Solve for v_x and i_x when $R_1 = R_2 = R_3 = R_4 = R_5 = R_6 = 10 \text{ k}\Omega$ and $v_S = 24 \text{ V}$.

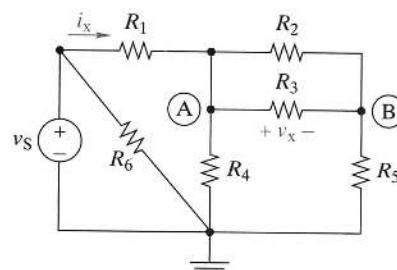


FIGURE P3-6

- 3-7 (a) Formulate node-voltage equations for the circuit in Figure P3-7.

- (b) Use MATLAB to find symbolic expressions for the node voltages in terms of the parameters in the circuit.

- (c) Find numeric values for v_A , v_B , and v_C when $R_1 = 1.5 \text{ k}\Omega$, $R_2 = 2.2 \text{ k}\Omega$, $R_3 = 3.3 \text{ k}\Omega$, $R_4 = 4.7 \text{ k}\Omega$, and $i_{S1} = i_{S2} = 2 \text{ mA}$.

- (d) Use OrCAD to verify that your solution to part (c) is correct.

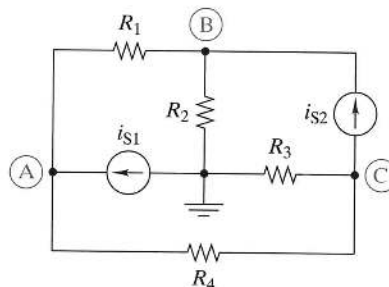


FIGURE P3-7

- 3-8 (a) Formulate node-voltage equations for the circuit in Figure P3-8.

- (b) Solve for v_x and i_x when $R_1 = R_4 = 1 \text{ k}\Omega$, $R_2 = R_3 = 250 \Omega$, $R_x = 500 \Omega$, and $v_S = 15 \text{ V}$.

- (c) Repeat part (b) for the case where R_4 is a variable resistor that varies from 10Ω to $10,000 \Omega$. At what value of R_4 is the voltage across R_x 0 V? (Hint: Use OrCAD to vary R_4 by trial and error to approach the answer.)

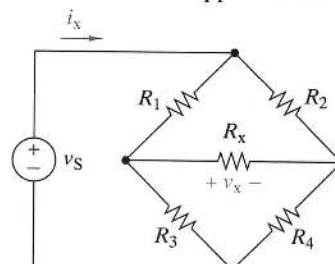


FIGURE P3-8

- 3-9 (a) Formulate mesh-current equations for the circuit in Figure P3-9.

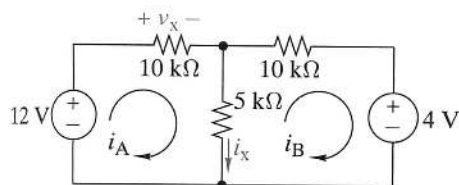


FIGURE P3-9

- (b) Solve for i_A and i_B .
 (c) Use these results to find v_x and i_x .
- 3-10 (a) Formulate mesh-current equations for the circuit in Figure P3-10.

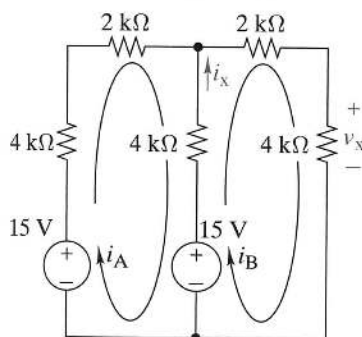


FIGURE P3-10

- (b) Solve for i_A and i_B .
 (c) Use these results to find v_x and i_x .
- 3-11 (a) Formulate mesh-current equations for the circuit in Figure P3-11.

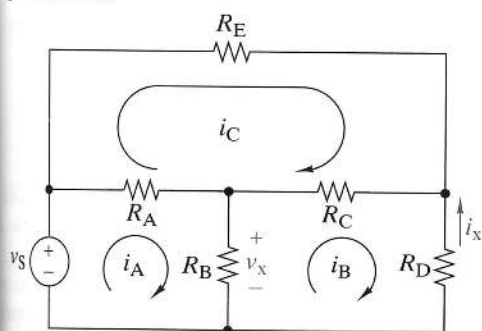


FIGURE P3-11

- (b) Formulate node-voltage equations for the circuit in Figure P3-11.

- (c) Which set of equations would be easier to solve? Why?
 (d) Using MATLAB, find v_x and i_x in terms of the mesh-current variables.
 (e) Using MATLAB, find v_x and i_x in terms of the node-voltage variables.

- 3-12 (a) Formulate mesh-current equations for the circuit in Figure P3-12.

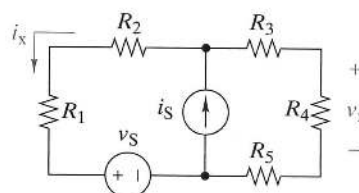


FIGURE P3-12

- (b) Solve for v_x and i_x when $R_1 = 200 \Omega$, $R_2 = 300 \Omega$, $R_3 = 50 \Omega$, $R_4 = 250 \Omega$, $R_5 = 200 \Omega$, $i_S = 100 \text{ mA}$, and $v_S = 25 \text{ V}$.
 (c) Find the total power dissipated in the circuit.

- 3-13 (a) Formulate mesh-current equations for the circuit in Figure P3-13.

- (b) Use MATLAB to find symbolic expressions for v_x and i_x in terms of the parameters in the circuit.
 (c) Find numeric values for v_x and i_x when $R_1 = R_2 = 4.7 \text{ k}\Omega$, $R_3 = 2.2 \text{ k}\Omega$, $R_4 = 1 \text{ k}\Omega$, $i_S = 3 \text{ mA}$, $v_{S1} = 10 \text{ V}$, and $v_{S2} = 2 \text{ V}$.
 (d) Find the power supplied by v_{S1} .
 (e) Use OrCAD to verify that your solutions to parts (c) and (d) are correct.

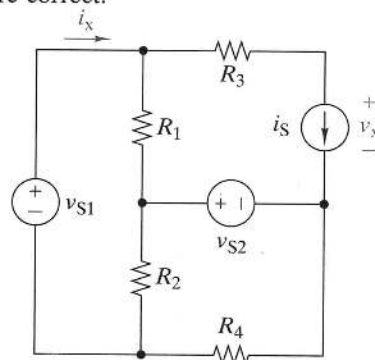


FIGURE P3-13

- 3-14 The circuit in Figure P3-14 seems to require two supermeshes since both current sources appear in two meshes. However, sometimes rearranging the circuit diagram will eliminate the need for a supermesh.

- (a) Show that supermeshes can be avoided in Figure P3-14 by rearranging the connection of resistor R_6 .
 (b) Formulate mesh-current equations for the modified circuit as redrawn in part (a).
 (c) Solve for v_x when $R_1 = R_2 = R_3 = R_4 = 2 \text{ k}\Omega$, $R_5 = R_6 = 1 \text{ k}\Omega$, $i_{S1} = 40 \text{ mA}$, and $i_{S2} = 20 \text{ mA}$.

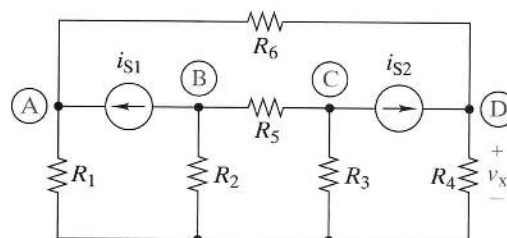


FIGURE P3-14

- (c) Find a symbolic expression for the equivalent resistance of the circuit by combining resistors in series and parallel. Compare your answer to the results from part (b).
-  **P3-19** Use OrCAD to find the node voltages v_A and v_B and the mesh currents i_A , i_B , and i_C in Figure P3-19.

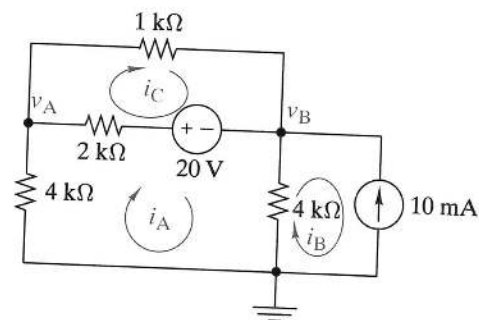


FIGURE P3-19

-  **3-20(a)** Use MATLAB and mesh-current analysis to solve for the mesh currents i_A , i_B , i_C , and i_D in Figure P3-20.
- (b)** Solve for the same currents using node-voltage analysis and compare the effort required with each technique.
- (c)** Use OrCAD to verify your results in the previous two parts.

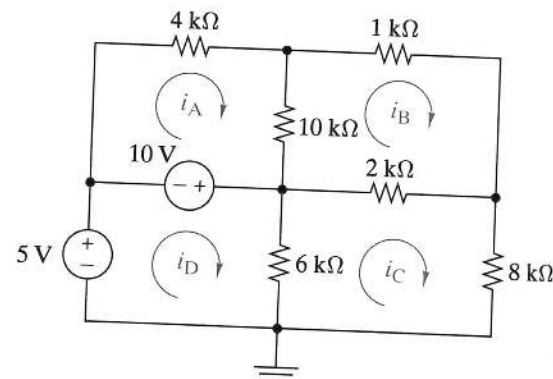


FIGURE P3-20

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OBJECTIVE 3-2 LINEARITY PROPERTIES (SECT. 3-3)

- Given a circuit containing linear resistors and one independent source, use the proportionality principle to find selected signal variables.
- Given a circuit containing linear resistors and two or more independent sources, use the superposition principle to find selected signal variables.

See Examples 3-10, 3-11, 3-12 and Exercises 3-18, 3-19, 3-20, 3-21, 3-22

3-15(a) Formulate mesh-current equations for the circuit in Figure P3-15.

(b) Formulate node-voltage equations for the circuit in Figure P3-15.

(c) Which set of equations would be easier to solve? Why?

(d) Find v_x and i_x using whichever method you prefer.

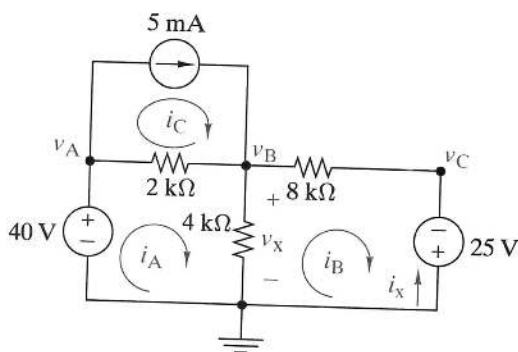


FIGURE P3-15

3-16 Use mesh-current equations, node-voltage equations, or simple engineering intuition to find the input resistance of the circuit in Figure P3-16.

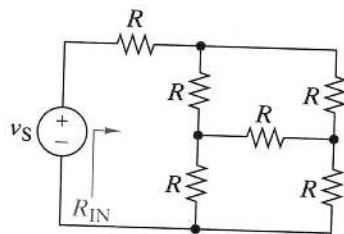


FIGURE P3-16

3-17 In Figure P3-17 all of the resistors are $1\text{ k}\Omega$ and $v_S = 10\text{ V}$. The voltage at node C is found to be $v_C = -2\text{ V}$ when node B is connected to ground. Find the node voltages v_A and v_D , and the mesh currents i_A and i_B .

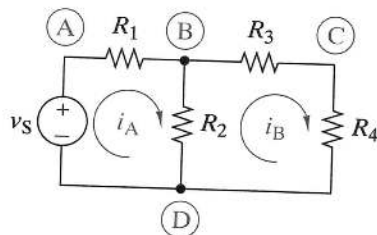


FIGURE P3-17

3-18 Use Figure P3-17 and MATLAB to solve the following problems:

(a) Using mesh-current analysis, find a symbolic expression for i_A in terms of the circuit parameters.

(b) Compute the ratio v_S/i_A .

(c) Find a symbolic expression for the equivalent resistance of the circuit by combining resistors in series and parallel. Compare your answer to the results from part (b).

3-19 Use OrCAD to find the node voltages v_A and v_B and the mesh currents i_A , i_B , and i_C in Figure P3-19.

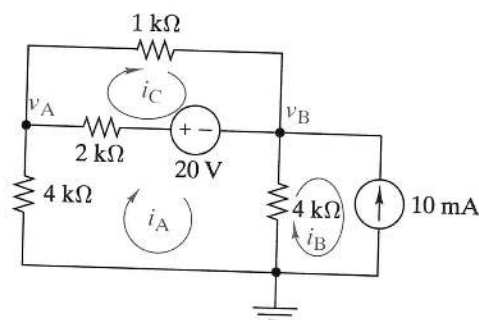


FIGURE P3-19

3-20(a) Use MATLAB and mesh-current analysis to solve for the mesh currents i_A , i_B , i_C , and i_D in Figure P3-20.

(b) Solve for the same currents using node-voltage analysis and compare the effort required with each technique.

(c) Use OrCAD to verify your results in the previous two parts.

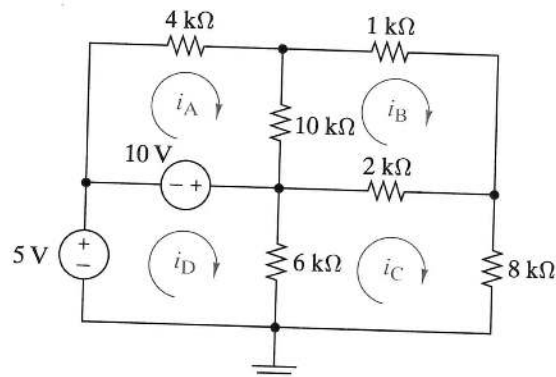


FIGURE P3-20

OBJECTIVE 3-2 LINEARITY PROPERTIES (SECT. 3-3)

(a) Given a circuit containing linear resistors and one independent source, use the proportionality principle to find selected signal variables.

(b) Given a circuit containing linear resistors and two or more independent sources, use the superposition principle to find selected signal variables.

See Examples 3-10, 3-11, 3-12 and Exercises 3-18, 3-19, 3-20, 3-21, 3-22